



can be achieved only by using a good-resolution sensor (detector)," says Kalidas Bhangare, managing director, Testo India. Housed within the bolometer, a sensor is the heart of a thermal imager. It has undergone several developments in recent years.

There are sensors that could integrate sensing, signal extraction, processing and comprehension. Under the encouragement of the USA's strategic defence initiative, such smart sensors began to appear.

There are two main types of smart sensors. The vision chips while used in the visible range, using smart sensing techniques, allow pre-processing due to the increase in growth of integrated microcircuitry. The other technology is more oriented to a specific use and fulfils its pre-processing goal through its design and structure.

Better FOV and focus

Comparing two notional thermal cameras that use the same un-cooled VOx detectors and 35mm lenses, we get the following estimates. The 320-pixel camera

How a thermographic camera works

A thermal imaging camera captures IR radiation, also known as blackbody radiation, emitted by an object. A specialised camera can detect this radiation just like a regular camera can detect light. Since ambient light levels do not matter, this specialised camera can be employed under total darkness, making it useful in underground and smoky places.

Major components of a thermal imaging camera are:

- An optics system
- Detector
- Amplifier
- Signal processing
- Display

These parts work together to render IR radiation given off by warm objects or flames into a visible light representation in real-time. For some specific cases, heat-resistant materials with waterproof housing are also employed.

A good thing with such cameras is that, for once looks do not matter. Two similar objects with similar temperature levels are displayed in the same colour, because the IR differentials are negligible. Many thermal imaging cameras use grayscale to represent normal temperature objects but highlight dangerously-hot surfaces in different colours.

will have a $13^\circ \times 10^\circ$ field of view (FOV), while the 640-pixel camera will have an $18^\circ \times 14^\circ$ FOV. This will give you a nearly 40 per cent increase in coverage area while still being able to detect a person from nearly 50 per cent farther away. The tactical and economic benefits of increased resolution are measurable and undeniable.

Fluke Ti450 IR camera delivers images automatically focused throughout the FOV. Using a multi-sharp focus technology, it takes multiple images and combines these to produce one in-focus image. Similar technologies allow clarity in thermal imaging cameras, which was missing earlier. With further wireless connectivity, these are handy in the field.

There is also a super-resolution technology from Testo that provides images suitable for data analytics. It works by increasing the instantaneous FOV and is not like a digital enhancement, explains Bhangare.

Adding such features drastically changes the imaging game. A thermal camera employs sensors that capture IR light radiating off objects. These require significant power.

Additionally, wireless capabilities like wireless connectivity also put a load on power requirements. This renders Keysight 5855A capable of working for about three hours at a stretch. An additional battery is provided to reduce load as much as possible.

Implementing the equipment

From sports like cricket to research sciences like astronomy, all fields employ thermal imaging cameras for different purposes. Maintenance, repair and operations people use these to locate over-heating joints and parts in order to eliminate potential hazards. Construction technicians check for heat leaks to improve efficiencies of air-conditioning systems in buildings using thermal imaging cameras.

Thermal imaging cameras have become so popular that Philippe Baylaucq even directed a 15-minute short dance film, Ora, entirely in 3D thermography. Bhangare explains customer demand as, "an imager that can see far off ob-



Fig. 2: Testo869 electrical maintenance